

Gavin J. Qu

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Education

University of Essex | November 2024 **MSc in Economics**

Honors: Distinction (~4.0 GPA)

Relevant coursework: Public Policy Evaluation (*Angrist & Pischke*), Data Science Economics (*ISLP/ISLR*), Graduate Econometric Methods (*Wooldridge*), Mathematical Methods (*Simon & Blume*), Advanced Microeconomic Theory (*Varian*), Behavioral Economics, Advanced Macroeconomic Theory (*Romer*).

University of Washington | June 2022 **BA in Economic & BA in Philosophy**

Honors: Dean's List (2020-2022)

Relevant coursework: Calculus I-III, Differential Equations, Statistics, Market Design, Econometrics, Public Economics, Environmental Economics, Intermediate Micro/Macro, Industrial Organization.

UIUC | 2025-Present **Math Preparation**

MATH 416: Abstract Linear Algebra (*Friedberg et al.*); MATH 314: Intro to Proof; MATH 447 Real Analysis (in-progress).

Certificates: Data Structures & Algorithms; Object-Oriented Programming in Java; CS50.

Research and Projects

MSc Thesis | University of Essex | Advisor: David Zentler-Munro

"From Medical Charts to Market Forces: How Health Types Shape Labour Market Outcomes" ([GitHub Link](#))

- Applied unsupervised machine learning techniques to the UKHLS (Understanding Society) Panel Data (2009–2022) using Python.
- Identified health-type heterogeneity using k-means clustering and modeled its specific effects on employment, earnings, and mortality dynamics.
- Utilized econometric modeling to analyze health-based disparities in the labor market.
- Implemented data science techniques to handle large-scale longitudinal data, ensuring results were statistically robust.

Research Communication & Open-Source

- **Creator, [econ-phd-advice](#) (open-source guide):** Built and maintain a community-driven Quarto/GitHub resource hub for economics PhD and predoc applicants.
- **Substack & Blog on Economics and Statistics.**

Experience

Pre-doctoral and PhD preparation

May 2026 – Present

Costco Travel

May 2025 – May 2026

- Conducted research-style exploratory analysis to identify drivers of operational variance; produced clear summaries for decision-making.
- Implemented statistical/econometric models (e.g., logistic regression) to quantify relationships and generate interpretable effect-based diagnostics.
- Applied machine learning methods (Random Forest, Gradient Boosting, NLP) to improve prediction and compare performance against baseline statistical models.
- Engineered reproducible Python/SQL pipelines to clean, merge, and validate large operational datasets (schema checks, missingness, outliers), improving data integrity and reducing manual work.
- Developed binary-classification models for price-variance detection (92% F1 score) with cross-validation.
- Standardized documentation and versioned workflows (LaTeX, Git) and automated ETL with Python/Bash to support scale and workflow reproducibility.

- **Technical Leadership:** Designed and hosted departmental workshops on SQL, statistical thinking, and causal inference.

Costco Corporate, Issaquah, WA

2022 — 2023

- Conducted forecasting and EDA to identify operational inefficiencies.

Math Instructor | Juni Learning, Seattle, WA

2021 — 2022

- Taught mathematical problem-solving and analytical reasoning from Pre-Algebra through AP Calculus.
- Assisted in designing curriculum on Economics and Personal Finance.

Technical Skills & Research Interests

Field of Interests: Labor, Health, Behavioral, Economics of Innovation.

Methodological Interests: Causal Inference, Bayesian, and Machine Learning Methods.

Causal Inference & Econometrics: Difference-in-Differences (DiD), Regression Discontinuity Design (RDD), Instrumental Variables (2SLS/LATE), Propensity Score Matching, Randomized Controlled Trials (RCTs), Panel Data Analysis, Quantile Regression, Causal Forest, Double-debiased ML.

Machine Learning: Regularization (Lasso/Ridge), Ensemble Methods (Random Forest, Gradient Boosting), Dimensionality Reduction (PCA/PCR), Generalized Additive Models (GAMs), Classification (LDA/QDA), Neural Networks (CNN/RNN), Cross-Validation & Bootstrapping, Q-Learning.

Mathematical Methods: Dynamic Programming, Constrained & Unconstrained Optimization, Differential & Difference Equations.

Programming & Tools – Python, R, Julia, SQL, Java, Git, *LaTeX*, Bash, Tableau, Claude Code.